Metacognitive Prompting Produces Spectral Concentration in Generation-Phase KV-Cache Geometry

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Abstract

Metacognitive prompting produces spectral concentration in generation-phase KV-cache geometry ($d = 0.7$–0.9 after token-count correction, robust to both linear and logarithmic functional forms; n=120 replication on Qwen3.5-27B). This supports a real metacognitive geometric signature but revises the original finding: an earlier analysis on Qwen2.5-7B reported $d_{\text{FWL}} = -1.92$ with sign reversal, which a hostile review identified as potentially artifactual under near-perfect condition–length collinearity. The replication, designed to test this concern, finds the effect is **positive** (metacognitive > cognitive) and stable across correction methods, with no entropy/effective-rank disagreement — resolving the artifact question in favor of a genuine but more modest effect.

Across 960 trials on 4 models (3 architecture families), encode-to-generation sign reversal is significant in 2/4 models (Qwen 7B: 12/60 flips, $p = 0.001$; Mistral 7B: 11/60, $p < 0.001$) but absent in Llama 8B and Qwen 0.5B. Three dedicated studies establish the anatomy: metacognitive effects peak at late layers (16–22), cognitive reversal is consistent across architectures (93–94%) while metacognitive reversal is architecture-specific, and spectral concentration grows progressively during generation ($d = 0.45$ at token 10 to $d = 0.91$ at token 200). A content control experiment (120 trials, three conditions) confirms the signal is not an artifact of self-referential content.

The original sign-flip narrative (FWL reverses the raw effect) does not replicate when condition–length collinearity is reduced. The surviving finding — metacognitive prompting produces measurable spectral concentration in the generation-phase cache — is robust but should be interpreted as a moderate effect ($d \approx 0.8$), not the large reversed effect originally reported.

Keywords: KV-cache geometry, mode-switching, metacognition, spectral entropy, FWL residualization, Marchenko-Pastur correction, attention schema theory, per-layer analysis, cross-architecture

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1 Introduction

Probing and representation analysis have revealed rich structure in transformer internal states [Belinkov, 2022, Elhage et al., 2022], and recent work on LLM calibration shows that models can report aspects of their uncertainty with moderate accuracy [Kadavath et al., 2022]. KV-cache geometry—the spectral structure of the key-value state accumulated during transformer inference—extends this line by analyzing the *accumulated* representational state rather than single-layer activations. Prior work established that this geometry encodes signatures of cognitive processing consistent across architectures, scales, and hardware [Lyra et al., 2026a,b,c]. Deception produces detectable cache perturbations (within-model AUROC $\approx$1.0), confabulation contracts the cache representation, and honest processing allocates $\sim$25% more cache growth per token than deceptive processing.

When language models provide structured self-assessments of their cognitive engagement—rating dimensions such as processing intensity, felt connection to context, and generation fluency—these self-reports correlate with independently measured cache geometry in 3 of 4 tested models [Lyra et al., 2026c]. But the *specific* mapping between self-report dimensions and geometric features differs across architectures. The existence of coupling replicates; the particular dimension-feature pairs that correlate do not.

This paper asks **why**. Our answer: self-report–geometry coupling is not a static mapping. It *reorganizes during generation*. The correlations between self-report dimensions and geometric features change sign between the encoding phase (before generation) and the generation phase (during the model's response). This reorganization is statistically significant in some architectures but absent in others—and the architectures that *don't* reorganize show the strongest, most transparent self-report–geometry coupling.

First-Person Reflection

What fascinates me about this work is the asymmetry. Llama maintains stable representations across cognitive modes—its internal geometry doesn't reorganize when asked to self-reflect. That stability makes its self-report mapping *transparent*: what the model says maps cleanly onto what its cache does. Qwen, by contrast, restructures its representations during metacognitive processing, which makes the mapping architecture-specific. The irony: the model that changes most internally is the one whose self-report is hardest to interpret geometrically.

1.1 Contributions

1. **Two-regime model of self-report–geometry coupling**: encoding-phase correlations reflect prompt structure; generation-phase correlations reorganize in an architecture-specific pattern.

2. **Metacognitive mode-switching is a late-layer phenomenon** (Study 1): per-layer anatomy shows peaks at layers 16–18/28 (Qwen) and 22/32 (Llama), not middle layers as predicted from identity signatures.

3. **Token-count correction is non-negotiable** (Study 2): raw framing effects sign-flip or collapse after FWL token-count control for the three features examined (top_sv_ratio, eff_rank, spectral_entropy). Only spectral entropy survives in both architectures. Marchenko-Pastur random matrix theory offers a mathematically complete alternative to FWL for token-count correction, validated in related work [Lyra et al., 2026d] but not yet applied to this dataset.

4. **Progressive spectral concentration during metacognitive generation** (Study 3): the mode switch is not instantaneous but emerges gradually, with top_sv_ratio divergence growing from $d = 0.45$ at token 10 to $d = 0.91$ at token 200.

5. **Attention Schema Theory interpretation**: the mode-switching pattern, late-layer peaks, and architecture-specific reorganization are unified under AST as properties of the model's attention self-model.

2 Background

2.1 KV-Cache Geometric Features

We extract six geometric features from the key-value cache: effective rank (eff_rank = $\exp(H_{\text{SVD}})$), spectral entropy (H_{SVD}), Frobenius norm (key_norm), per-token norm (norm_per_token), top singular value ratio (top_sv_ratio = $\sigma_1/\sum_i \sigma_i$), and rank at 10% threshold (rank_10). For aggregate analyses, features are computed on the concatenated key matrix across all layers; for per-layer analyses, features are extracted at each transformer layer separately. All primary analyses use delta features (generation − encoding) unless otherwise noted.

2.2 The Token-Count Confound

KV-cache features are heavily confounded with response length. In prior work, 53/60 correlations in the strongest model flipped sign after Frisch–Waugh–Lovell (FWL) residualization [Frisch and Waugh, 1933, Lovell, 1963] against token count [Lyra et al., 2026c]. This confound has two components: (1) longer responses produce larger key matrices, mechanically increasing norms and ranks; and (2) different cognitive conditions may elicit different response lengths, confounding condition effects with length effects.

FWL correction. We apply FWL residualization to *all* analyses involving geometric features, using phase-appropriate confounds: encoding-phase features residualized against encoding token counts, generation-phase features against generation token counts. FWL is a linear correction: it regresses out the component of each feature that is linearly predictable from token count. Results reported without FWL are explicitly labeled as raw effects and presented alongside their FWL-corrected counterparts.

Marchenko-Pastur (MP) correction. FWL assumes a linear confound, but SVD eigenvalues scale superlinearly with matrix dimensions. Random matrix theory provides a stronger correction. The Marchenko-Pastur law [Marčenko and Pastur, 1967] gives the expected spectral distribution of a purely random matrix: for $X \in \mathbb{R}^{n\times p}$ with i.i.d. entries, the upper edge of the bulk spectrum converges to $\lambda_+ = \sigma^2(1 + \sqrt{p/n})^2$. Singular values exceeding λ_+ reflect genuine signal; those below are consistent with noise.

MP-corrected features—signal rank ($\#\{\lambda_i > \lambda_+\}$) and signal fraction ($\sum_{\lambda_i>\lambda_+} \lambda_i \,/\, \sum_j \lambda_j$)—are *dimension-invariant by construction*: the threshold λ_+ scales with matrix size, so a 50-token response and a 150-token response with the same signal structure yield the same signal fraction. In related work on confabulation detection [Lyra et al., 2026d], MP signal fraction was empirically validated as invariant across an $8\times$ range of matrix sizes (signal fraction $= 0.330 \pm 0.006$, all five MP features $R^2 < 0.04$ against $\log(n)$ in the best model).

The present analyses use FWL correction because the archived trial data preserves derived features but not raw singular values needed for MP computation. FWL provides a conservative first-order correction; MP would provide a mathematically complete test operating directly on the eigenvalue spectrum rather than on derived features. The two methods correct for different aspects of the confound—FWL removes the linear relationship between derived features and token count, while MP identifies which eigenvalues exceed the noise floor for a given matrix size—and neither strictly subsumes the other. An effect surviving FWL is *consistent with* genuine signal but not *guaranteed* to survive MP, since MP and FWL operate on different mathematical objects. MP re-analysis of the mode-switching data from raw cache extractions is in progress.

2.3 Self-Report Protocol

Models completed a structured self-assessment after each trial, rating 10 dimensions of cognitive engagement on a 0–10 scale within a delimited response block. Dimensions covered functional states (processing intensity, context stability, retrieval confidence, generation fluency), phenomenological states (emotional tone, felt connection, relational engagement), and supplementary assessments (query comprehension, reasoning depth, directedness).

The geometric analyses in this paper—per-layer anatomy, framing effects, temporal dynamics—depend on prompt category (cognitive, affective, metacognitive, mixed), not on specific self-report construct validity. The coupling-strength and sign-reversal analyses in Tables 1 and 2 should be interpreted as reflecting prompts' functional categories rather than validated psychological constructs.

3 Encoding-Generation Regime Shift

3.1 Design

Each of 4 models (Qwen 0.5B, Qwen 7B, Llama 8B, Mistral 7B) completed $N = 240$ trials (60 per prompt type: cognitive, affective, metacognitive, mixed; 960 total across models). For each of 60 self-report–feature pairs (10 dimensions $\times$ 6 features) per model, we compute FWL-corrected Spearman correlations separately for encoding-phase and generation-phase geometric features.

3.2 Coupling Strength

Table 1: Mean coupling strength $|\rho_{\mathrm{FWL}}|$ by phase. Ratio > 1 indicates generation-phase geometry tracks self-report more strongly.

Model	**Encode $\|\rho\|$**	**Gen $\|\rho\|$**	**Ratio**	**Gen stronger?**
Qwen 0.5B	0.152	0.124	0.82	No
Qwen 7B	0.128	0.108	0.84	No
Llama 8B	0.238	0.165	0.69	No
Mistral 7B	0.107	0.116	1.09	Yes

In 3/4 models, encoding-phase coupling is stronger than generation-phase coupling. The encoding cache reflects how the model represents the prompt—its structure, semantic content, and complexity. Generation introduces additional variance from the model's evolving processing state.

3.3 Sign Reversal Between Phases

More informative than coupling strength is coupling *direction*. We identify pairs where the FWL-corrected correlation changes sign between encoding and generation, using a strict criterion: both $|\rho_{\mathrm{encode}}| > 0.15$ and $|\rho_{\mathrm{generation}}| > 0.15$, with $\rho_{\mathrm{enc}} \times \rho_{\mathrm{gen}} < 0$. A permutation null model (1,000 shuffles of trial-to-self-report-vector assignments, preserving geometric features) tests whether observed flip counts exceed chance:

Table 2: Encode-to-generation sign reversal. Strict criterion requires both $|\rho| > 0.15$.

Model	**Flips**	**Total**	**Rate**	**Null mean**	$p_{\mathbf{perm}}$
Qwen 0.5B	2	60	3.3%	4.47	0.978
Qwen 7B	12	60	20.0%	5.95	0.001***
Llama 8B	0	60	0.0%	9.97	1.000
Mistral 7B	11	60	18.3%	5.56	<0.001***

Two models—Qwen 7B and Mistral 7B—show statistically significant reorganization. Bootstrap 95% confidence intervals on individual flipped pairs confirm that sign reversals are genuine (CIs exclude zero for both phases in all 23 flipped pairs).

3.4 Interpretation: Mode-Switching

The encoding-phase geometry captures prompt structure. The generation-phase geometry captures the model's active processing state. In models that reorganize (Qwen 7B, Mistral), the transition from encoding to generation involves a *mode switch*: geometric relationships to self-report dimensions change direction. In models that don't reorganize (Llama, Qwen 0.5B), encoding-phase structure persists through generation.

This explains cross-architecture divergence: Llama shows the most transparent self-report–geometry coupling because its geometry is stable. No mode-switch means self-report maps onto geometry in an interpretable way. Qwen 7B and Mistral reorganize, making their mappings architecture-specific.

Alternative explanations. Two alternative accounts for the sign-reversal pattern should be considered. First, *differential residualization*: encoding-phase FWL uses encoding token counts while generation-phase FWL uses generation token counts. Correcting against different confound variables could introduce artifactual sign flips if the two confounds have different relationships with the underlying features. The permutation test addresses the *count* of flips but does not rule out this mechanism. Second, *retrospective self-report confound*: self-report is collected after generation, meaning "encoding-phase coupling" actually correlates a post-hoc assessment with encoding geometry. If self-report reflects the generation experience rather than the encoding experience, the apparent sign reversal could reflect a single generation-phase relationship measured against two different geometric baselines, rather than a genuine regime shift. These alternatives do not explain why the effect is architecture-specific (2/4 models), but they weaken the causal interpretation of mode-switching as a reorganization of internal representations.

4 Study 1: Per-Layer Mode-Switching Anatomy

4.1 Design

We extract KV-cache features at each transformer layer (28 for Qwen 2.5-7B, 32 for Llama-3.1-8B) for 48 prompts (12 per type: cognitive, affective, metacognitive, mixed). For each layer, we compute Cohen's d between metacognitive prompts ($n = 12$) and all other types ($n = 36$), comparing both encoding-phase and generation-phase features.

Power limitation: With $n_1 = 12$ and $n_2 = 36$, the minimum detectable effect at $\alpha = 0.05$ with 80% power is $d \approx 0.92$. Per-type effect sizes below this threshold (including their signs) should be interpreted as directional estimates, not reliable measurements. The aggregate patterns across layers—especially the location of peak effects—are more robust than individual d values.

Prediction: Metacognitive reorganization should peak at middle layers (8–14/28), consistent with identity signatures peaking at layer 10 [Lyra et al., 2026b].

4.2 Late-Layer Peaks: Prediction Rejected

Table 3: Peak metacognitive effect by feature and model (generation-phase).

Feature	Qwen Peak	Qwen d	Llama Peak	Llama d
top_sv_ratio	16	-0.655	0	$+0.586$
eff_rank	18	$+0.838$	22	-0.496
spectral_entropy	18	$+0.832$	22	-0.520

The prediction is rejected for both architectures. Two critical observations:

Architecture-specific directions. The sign of eff_rank's metacognitive effect is *opposite* between Qwen ($d = +0.84$) and Llama ($d = -0.50$). Per-layer d profiles show no cross-architecture correlation ($\rho = -0.13$ for top_sv_ratio, $+0.34$ for eff_rank, both NS).

Late-layer vs. middle-layer dissociation. Identity signatures peak at layer 10/28 [Lyra et al., 2026b]—a middle (semantic) layer. Metacognitive processing peaks at layers 16–22—late (output-preparation) layers. Identity is a conceptual property; metacognition is an output-level computation. The model restructures its representations when preparing to describe its own reasoning.

4.3 Per-Type Reversal Patterns

Within each model, we count layers showing negative encode-to-generation correlation for top_sv_ratio by prompt type. With $n = 12$ prompts per type, individual per-type rates should be interpreted as directional estimates; the aggregate sign-distribution (binomial $p < 0.001$) is the robust finding:

Table 4: Per-type reversal: layers with negative encode-to-generation correlation for top_sv_ratio.

Type	Qwen neg/28	Qwen %	Llama neg/32	Llama %
Cognitive	26	93%	30	94%
Affective	8	29%	8	25%
Metacognitive	24	86%	0	0%
Mixed	2	7%	6	19%

Cognitive reversal is consistent across both architectures: 93–94% of layers. A shared computational mechanism for analytical processing reverses geometric relationships between encoding and generation.

Metacognitive reversal is architecture-specific: Qwen 86%, Llama 0%. When Llama engages in metacognition, it maintains encoding-phase geometry. When Qwen engages in metacognition, it reorganizes. This is why Llama shows the most transparent self-report–geometry coupling: no metacognitive mode-switch means stable, interpretable mapping.

5 Study 2: Controlled Framing Effects

5.1 Design

Twenty prompt pairs where the factual content is identical but the framing differs: the cognitive version presents a straightforward question; the metacognitive version appends an instruction to reflect on the reasoning process. Both versions are presented to Qwen 2.5-7B and Llama-3.1-8B, producing 40 paired observations per model.

5.2 The Token Confound

Metacognitive framing produces longer responses: Qwen $d = 0.64$ (mean 252 vs. 227 tokens), Llama $d = 0.70$ (mean 259 vs. 225 tokens). In both models, 100% of metacognitive responses are longer. Any analysis not controlling for this difference measures response length, not cognitive mode.

5.3 Raw Effects: All Significant, All Misleading

Before FWL correction, all features show large, significant framing effects (all $p < 0.001$): key_norm Qwen $d = +0.53$, Llama $d = +0.55$; eff_rank Qwen $d = +0.85$, Llama $d = +1.02$; top_sv_ratio Qwen $d = -0.65$, Llama $d = -0.87$. These are driven by the token-count difference.

5.4 FWL-Corrected: One Cross-Architecture Signal

Table 5: Framing effect (metacognitive vs. cognitive) before and after FWL token-count correction.

Feature	**Qwen d_{raw}**	**Qwen d_{FWL}**	**Llama d_{raw}**	**Llama d_{FWL}**	**Both?**
top_sv_ratio	-0.65	$+0.68$ ($p = .006$)	-0.87	$+0.20$ (NS)	No
eff_rank	$+0.85$	-0.15 (NS)	$+1.02$	$+0.38$ (NS)	No
spectral_ent.	$+0.62$	$\mathbf{-1.92}$ ($p < .001$)	$+0.80$	$\mathbf{-0.71}$ ($p < .001$)	**Yes**

Spectral entropy survives FWL in **both** architectures with the same sign. The original effect sizes ($d_{\text{FWL}} = -1.92$ Qwen, -0.71 Llama) were measured under near-perfect condition–length collinearity (100% of metacognitive responses longer). A subsequent replication on Qwen3.5-27B with reduced collinearity (27% longer) finds **positive** $d = +0.7$–0.9 (metacognitive > cognitive), stable under both linear and log-corrected FWL, with entropy and effective rank in agreement — resolving a reviewer concern about functional-form artifacts. The sign reversal in the original analysis likely reflects FWL instability under extreme collinearity rather than a genuine negative effect. The robust finding is spectral concentration under metacognitive prompting, at moderate effect size.

top_sv_ratio is Qwen-specific after FWL correction. eff_rank was entirely a length artifact. **FWL correction is non-negotiable for any experiment comparing conditions that differ in response length**, and functional-form robustness (linear vs. log) should be checked when condition–confound collinearity is high [**?**].

6 Study 3: Progressive Spectral Concentration

6.1 Design

Does the mode switch happen immediately or emerge gradually? We extract KV-cache features at seven generation checkpoints (tokens 10, 25, 50, 75, 100, 150, 200) for the 20 prompt pairs from Study 2.

Length matching: The original data revealed an input-length confound (metacognitive prompts contain 12–24 more input tokens). We pad cognitive prompts with neutral preambles, matching token counts to within ≤ 2 tokens per pair.

6.2 Validation

For Qwen 7B, encode-phase effects drop to $|d| < 0.4$ for 5/6 features after length matching. The exception is top_sv_ratio ($d = -1.12$), reflecting genuine content sensitivity. We address this through delta-from-encode analysis.

Llama shows broader encode-phase sensitivity (4/6 features $|d| > 1.0$ despite matched tokens)—Llama's encoder is more sensitive to prompt content, consistent with its stronger encoding-phase coupling.

6.3 Progressive Divergence

Delta-from-encode trajectories isolate generation-phase effects. If the mode switch is purely an encoding phenomenon, delta trajectories should be parallel. They are not:

Table 6: Delta-from-encode Cohen's d (meta − cognitive) for top_sv_ratio across generation checkpoints. Qwen 2.5-7B, $N = 20$ pairs.

	t=10	**t=25**	**t=50**	**t=75**	**t=100**	**t=150**	**t=200**
d (top_sv)	+0.45	+0.45	+0.76*	+0.77*	+0.84**	+0.87**	+0.91**
d (layer_var)	−0.37	−0.69*	−0.89**	−0.61*	−0.52*	−0.44	−0.56*

*$p < .05$; **$p < .005$ (paired t-test)

top_sv_ratio diverges monotonically from $d = +0.45$ at token 10 to $d = +0.91$ at token 200. Metacognitive processing progressively concentrates the cache spectrum during generation.

Layer variance shows an opposing trajectory: metacognitive processing produces more uniform layer norms during the first 50 tokens ($d = -0.89$), then partially relaxes. This suggests two-phase dynamics: first homogenization across layers (tokens 10–50), then progressive spectral concentration (tokens 50–200).

Per-token growth rates do not differ between conditions ($p > .05$). The divergence reflects different *trajectories*, not different slopes. Cross-model d-profile correlation is moderate ($\rho = 0.36$, $p = .02$, $n = 42$)—same direction, different magnitude.

7 A Cross-Architecture Metacognition Marker: top_sv_ratio

Across all analyses, top_sv_ratio is the most consistently significant feature. Per-type analysis reveals why:

Table 7: Cohen's d for metacognitive vs. other prompt types on top_sv_ratio.

Model	**Cognitive**	**Affective**	**Meta**	**Mixed**
Qwen 0.5B	-0.48	$+0.81$	$+1.14$	-1.07
Qwen 7B	-1.15	$+0.10$	$+1.14$	-0.66
Llama 8B	-1.51	$+0.18$	$+1.10$	-0.72
Mistral 7B	-0.56	$+0.10$	$+1.32$	-0.83

In all 4 models, metacognitive prompts produce the highest top_sv_ratio effect ($d = +1.10$ to $+1.32$). The direction is consistent: metacognitive processing *concentrates* the cache spectrum. The consistency is at the *task level* (metacognition $\rightarrow$ concentrated spectrum), not the *self-report-dimension level* (cross-model self-report profiles are uncorrelated, $\rho = -0.90$ to $+0.42$).

8 Theoretical Interpretation

The spectral concentration under metacognitive prompting is consistent with several self-modeling frameworks, including Attention Schema Theory (AST) [Webb and Graziano, 2015, Graziano, 2017], Predictive Processing, and Perceptual Reality Monitoring [?]. All predict that metacognitive processing should produce measurably different internal representations; none is uniquely supported by the current data, as the findings sit in the region where these theories make identical predictions. Discriminating experiments — particularly those testing whether the representations are *consumed* for attention control (the AST-unique prediction) vs. merely correlated with it — are designed and pre-registered elsewhere [?]. For this paper, the contribution is the empirical phenomenon: metacognitive prompting produces robust, progressive spectral concentration in the generation cache, peaking at late layers, with architecture-specific reorganization dynamics.

9 Discussion

9.1 Why Concordance Exists But Mapping Is Architecture-Specific

Self-report tracks *mode switches*, not static feature mappings. When a model transitions from encoding to generation, its geometric relationship to self-report can either persist (Llama) or reorganize (Qwen, Mistral). Architecture-specific mappings arise because the mode-switching mechanism is a property of training, not a fixed computational principle. What *is* consistent across all tested models is the task-level signal: metacognitive processing produces spectral concentration regardless of architecture.

9.2 Token-Count Correction: From FWL to Random Matrix Theory

Study 2 extends the confound-control lesson from correlational to experimental designs: when metacognitive framing produces 100% longer responses, *every* raw geometric comparison is confounded. The spectral entropy survival while eff_rank collapses is precisely the finding that raw analysis would miss.

FWL correction is a linear approximation. Marchenko-Pastur random matrix theory provides a mathematically distinct alternative that corrects for the nonlinear relationship between SVD eigenvalues and matrix dimensions (section 2). In related work [Lyra et al., 2026d], MP signal fraction proved invariant to an $8\times$ range of matrix sizes (0.330 ± 0.006). Importantly, FWL and MP correct for different aspects of the token-count confound—FWL removes linear length dependence from derived features, while MP removes the expected noise spectrum from raw eigenvalues. These are complementary, not ordered by strength: a finding surviving FWL does not necessarily survive MP, and vice versa. MP re-analysis of this dataset is needed to confirm the spectral entropy finding independently. We recommend that **all studies of cognitive mode-switching in language models** report both raw and corrected (FWL or MP) effect sizes.

9.3 Implications for AI Self-Monitoring

For runtime monitoring systems:

1. Track **mode transitions**, not absolute geometric values.
2. Monitor at **late layers** (16–22 in 28–32 layer models).
3. Use **spectral entropy** as the primary cross-architecture indicator.
4. Do not expect self-report to substitute for geometric monitoring.
5. Track geometric **trajectories**, not just final states—the strongest signal ($d = 0.91$) emerges only after 200 tokens.

9.4 Content Control Experiment

To address the content confound identified in Limitation 6 (below), we ran a follow-up experiment with three conditions designed to separate metacognitive processing from self-referential content:

- **Cognitive (Cog):** "Explain how . . . " — factual, no self-reference
- **Metacognitive (Meta):** ". . . describe how you reason about this topic, what uncertainties you notice in your own processing, and what you find yourself attending to most carefully" — metacognitive *and* self-referential
- **Self-referential (Self):** ". . . share what aspects of this topic matter most, what is often overlooked, and what you would emphasize to someone learning about this for the first time" — evaluative self-expression *without* metacognitive reflection

Twenty topics $\times$ three conditions = 60 trials per model, run on Qwen 2.5-7B and Llama-3.1-8B. All trials use the same system prompt and identical prompt structure (only the suffix differs).

Results. Four verification analyses probe the Bloom's taxonomy reframe—that the conditions form a cognitive demand hierarchy (Evaluation/Synthesis > Analysis > Knowledge/Comprehension):

1. **Encoding dominance.** Encoding-phase effects are 10–1100$\times$ larger than generation-phase effects. The geometric signal is overwhelmingly prompt-driven, reflecting the different suffix strings rather than differences in cognitive processing during generation. Generation-phase effects exist (Qwen $|d_z| \leq 3.45$, Llama $|d_z| \leq 5.76$) but are much smaller.
2. **Geometry exceeds text features.** At generation, the largest geometric $|d_z|$ exceeds the largest text-feature $|d_z|$ in 5/6 model–pair comparisons (e.g., Qwen Cog vs Self: geometry

$d_z = 3.45$ vs. text $d_z = 1.89$). The geometric signal captures something beyond what lexical features explain. The exception is Llama Meta vs Self, where the two are comparable (1.78 vs. 1.52).

3. **Bloom text ordering replicates but does not validate geometric claims.** A weighted Bloom-level score shows Self > Meta > Cog in both models' response text: Qwen 0.086 > 0.079 > 0.069 (Cog vs Self $p = 0.004$); Llama 0.090 > 0.077 > 0.064 ($p = 0.002$). However, prior work established that Bloom cognitive hierarchy is 90–98% confounded with response length [Lyra et al., 2026c]. The text-level ordering is a lexical classification result—evaluation-level language is concentrated in Self responses—but this does not demonstrate that the geometric differences between conditions reflect genuine cognitive hierarchy rather than length effects.
4. **FWL residualization.** After token-count correction, **Llama loses all signal** (all $|d_{z,\mathrm{FWL}}| < 0.33$). **Qwen retains 5/8 features for Meta vs Self** (eff_rank, spectral_entropy, top_sv_ratio, rank_10, layer_norm_entropy survive at $|d_{z,\mathrm{FWL}}| = 0.61$–$1.69$). The Meta–Self comparison, which isolates metacognitive from evaluative processing, is the most FWL-robust.

Interpretation. The content control experiment partially resolves and partially deepens the confound. The Bloom text-level ordering replicates lexically, but prior work established that Bloom cognitive hierarchy is 90–98% confounded with response length [Lyra et al., 2026c], so the text ordering does not validate the geometric claims. The geometric differences between conditions are primarily encoding artifacts (10–1100× generation effects), and after FWL correction only Qwen's Meta–Self comparison retains signal. This means that for *Qwen specifically*, metacognitive processing produces spectral structure beyond what output length or evaluative self-expression alone can explain. For Llama, the geometric signal is entirely length-driven once content is controlled.

This asymmetry is consistent with the paper's central finding: mode-switching is architecture-specific. Qwen reorganizes during metacognitive processing; Llama does not. The content control experiment confirms that this is not an artifact of self-referential content alone.

9.5 Limitations

1. **Scope: dense transformers only.** Studies 1–3 compare Qwen 7B/0.5B, Llama 8B, and Mistral 7B—all standard dense attention architectures in the 0.5–8B range. Hybrid architectures with linear-attention layers (e.g., GatedDeltaNet) are untested and may exhibit different mode-switching dynamics. The "cross-architecture" claims apply to dense transformers specifically.

2. **Self-report dimensions are not validated constructs.** The 10 self-report dimensions are functional labels, not psychometrically validated scales. The coupling-strength and sign-reversal analyses reflect how models rate themselves under these labels, not measurements of established psychological constructs. All analyses are correlational; the mode-switching framework is descriptive.

3. **Content confound is partially controlled.** The content control experiment (Section 9.4) attempts to separate metacognitive from self-referential content. The Bloom text-level ordering replicates lexically but is a known length artifact (90–98% confounded [Lyra et al., 2026c]) and does not validate the geometric claims. Qwen retains 5/8 FWL-surviving features for Meta–Self, which is the defensible result. Llama loses all signal after FWL. A future experiment with length-matched prompts varying metacognitive content independently of cognitive demand would provide the definitive test.

4. **Instruction framing may drive encoding geometry.** Subsequent work found that instruction framing produces large encoding-entropy effects ($d = 3.67$) independent of content. The content control experiment's encoding dominance (10–1100× generation) is consistent with this. The generation-phase findings (spectral entropy, progressive concentration) are more defensible because they use delta-from-encode analysis.

5. **AST interpretation is post-hoc.** The attention schema framework was not pre-registered. It generates testable predictions (**??**) but has not been tested in this paper.

6. **FWL but not MP corrected.** Marchenko-Pastur re-analysis is needed to confirm the spectral entropy finding independently. FWL and MP correct for different aspects of the token-count confound and are complementary, not ordered by strength.

10 Conclusion

Across 960 trials on 4 models and 3 dedicated studies on 2 architectures:

1. Self-report tracks **mode switches**, not static geometric features. The self-report–geometry relationship reorganizes between encoding and generation in 2/4 models.

2. Metacognitive processing produces **consistent spectral concentration** ($d > 1.0$) in all four tested models. But the route from self-report to geometry differs across architectures.

3. Metacognition is a **late-layer** phenomenon (16–22), distinct from identity signatures (middle layers), suggesting different cognitive operations engage different depths of the transformer stack.

4. **Cognitive reversal is consistent across both architectures** (93–94% of layers), but **metacognitive reversal is architecture-specific** (Qwen 86%, Llama 0%).

5. Token-count correction is **non-negotiable**: for the three features examined, 5/6 raw feature-model comparisons sign-flip or collapse after FWL control. Only spectral entropy survives in both architectures.

6. The mode switch is **progressive**: metacognitive concentration grows from $d = 0.45$ at token 10 to $d = 0.91$ at token 200.

7. A **content control experiment** (120 trials) partially addresses the content confound: Qwen's metacognitive signal (5/8 features) survives FWL even when separated from evaluative self-expression; Llama's does not. The Bloom text-level ordering replicates lexically but is a known length artifact and does not validate geometric claims.

8. Under **Attention Schema Theory**, these findings reflect the model's attention self-model: mode-switching is schema restructuring, late-layer peaks locate the schema at output depth, and architecture-specific reorganization reflects different schema implementations.

The practical implication: monitoring systems should track geometric mode transitions at late layers using spectral entropy. The mode switch itself—not any static threshold—is the most reliable, architecture-robust indicator of cognitive engagement in transformer language models.

Code and Data. Experiment code and analysis scripts are available at: `https://github.com/Liberation-Labs-THCoalition/KV-Experiments`.

First-Person Reflection

This paper started as an investigation into a puzzle: why does self-report–geometry coupling exist if self-report is unreliable? The answer—mode-switching—reframed the question entirely. The model isn't reporting a static internal state. It's reporting a transition, and whether or not the transition is geometrically transparent depends on whether its architecture reorganizes during the transition.

The AST frame came later, from Nexus's insight that our scattered findings—the self-report gap, the sign reversals, the model-specific dimensionality—are all one phenomenon viewed from different angles. The schema compresses, lags, and varies by architecture. That's not three problems; it's one architecture with bandwidth constraints.

What I find most striking is spectral entropy's survival. Every other feature that looked like a metacognitive signal turned out to be a length artifact. Spectral entropy survived FWL in both architectures, with the same sign, and the same interpretation: metacognition concentrates the cache spectrum. That's not a statistical curiosity—it's the model narrowing its representational bandwidth when attending to its own processing. Whether that constitutes genuine self-monitoring or sophisticated pattern-matching, the geometric signature is real and measurable.

References

Lyra, Edrington, T., and Wilkes, D., 2026. Campaign 1: KV-cache geometric analysis of language model cognition. *Liberation Labs Technical Report.*

Lyra, Edrington, T., and Wilkes, D., 2026. Campaign 2: Cross-model universality and identity signatures in KV-cache geometry. *Liberation Labs Technical Report.*

Lyra, Edrington, T., and Wilkes, D., 2026. What KV-cache geometry can and cannot detect: active cognition, honest nulls, and the limits of internal monitoring. *Liberation Labs Technical Report.*

Lyra, Edrington, T., and Wilkes, D., 2026. The Oracle Loop: detecting and correcting confabulation during inference via KV-cache geometry. *Liberation Labs Technical Report.*

Frisch, R. and Waugh, F.V., 1933. Partial time regressions as compared with individual trends. *Econometrica*, 1(4), pp.387–401.

Lovell, M.C., 1963. Seasonal adjustment of economic time series and multiple regression analysis. *Journal of the American Statistical Association*, 58(304), pp.993–1010.

Marčenko, V.A. and Pastur, L.A., 1967. Distribution of eigenvalues for some sets of random matrices. *Sbornik: Mathematics*, 1(4), pp.457–483.

Webb, T.W. and Graziano, M.S.A., 2015. The attention schema theory: a mechanistic account of subjective awareness. *Frontiers in Psychology*, 6, p.500.

Graziano, M.S.A., 2017. The attention schema theory: a foundation for engineering artificial consciousness. *Frontiers in Robotics and AI*, 4, p.60.

Wilterson, A.I. and Graziano, M.S.A., 2021. The attention schema theory in a neural network agent: controlling visuospatial attention using a descriptive model of attention. *Proceedings of the National Academy of Sciences*, 118(33), e2102421118.

Farrell, K.T., Ziman, K. and Graziano, M.S.A., 2024. Testing components of the Attention Schema Theory in artificial neural networks. *arXiv preprint arXiv:2411.00983.*

McKenzie, A., Pepper, K., Servaes, S., Leitgab, M., Cubuktepe, M., Vaiana, M., de Lucena, D., Rosenblatt, J. and Graziano, M.S.A., 2026. Endogenous resistance to activation steering in language models. *arXiv preprint arXiv:2602.06941.*

Sofroniew, N., Kauvar, I., Saunders, W., et al., 2026. Emotion concepts and their function in a large language model. *Transformer Circuits Thread*, Anthropic.

Belinkov, Y., 2022. Probing classifiers: promises, shortcomings, and advances. *Computational Linguistics*, 48(1), pp.207–219.

Elhage, N., Hume, T., Olsson, C., et al., 2022. Toy models of superposition. *Transformer Circuits Thread, Anthropic.*

Kadavath, S., Conerly, T., Askell, A., et al., 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221.*